Notation. λ security parameter; U_n uniform on $\{0,1\}^n$; p.p.t. = probabilistic poly-time; negl = smaller than every $\frac{1}{\text{poly}}$ for large λ; $X \approx Y$ computational indistinguishability ($\forall$ p.t. D: $|\Pr[D(X)=1]-\Pr[D(Y)=1]|=\text{negl}$); $X \equiv Y$ identical distributions; $\oplus$ = bitwise xor; $\|$ = concatenation; $\langle x,y\rangle = \sum_i x_i y_i \bmod 2$; $F_k(\cdot)=F(k,\cdot)$.

1 · Encryption: syntax and perfect secrecy

Def 1.1 (encryption scheme). $\Pi=(\mathcal{M},\mathcal{C},\mathcal{K},\text{Gen},\text{Enc},\text{Dec})$: $\text{Gen}(1^\lambda)$ is randomized and samples $k\leftarrow\mathcal{K}$; $\text{Enc}:\mathcal{K}\times\mathcal{M}\to\mathcal{C}$ may be randomized; $\text{Dec}:\mathcal{K}\times\mathcal{C}\to\mathcal{M}$ is deterministic. **Correctness:** $\text{Dec}(k,\text{Enc}(k,m))=m$ for all k,m.

Def 1.2 (perfect secrecy, Shannon). For all $m_0\neq m_1$: $\text{Enc}(K,m_0)\equiv\text{Enc}(K,m_1)$, where $K\leftarrow\text{Gen}$. Equivalently $\Pr[M=m\mid C=c]=\Pr[M=m]$ for every c in the support — the ciphertext leaks nothing about the message.

Def 1.3 (semantic security). For every function f on $\mathcal{M}$, every distribution M, every adversary A with side information $g(M)$, there is S with

$$\Pr[A(c,g(M))=f(M)]\le\Pr[S(g(M))=f(M)].$$

Prop 1.4. Def 1.2 $\Leftrightarrow$ Def 1.3. ($\to$: $S(g(M)):=A(\text{Enc}(K,m_0),g(M))$. $\leftarrow$: if $\text{Enc}(K,m_0)\neq\text{Enc}(K,m_1)$, take M uniform on $\{m_0,m_1\}$, $g=\perp$ and a distinguisher A; then S is at best a random guess while A wins $>\frac12$.)

One-time pad (Ex 1.5). $\mathcal{K}=\mathcal{M}=\mathcal{C}=\{0,1\}^n$, $k\leftarrow U_n$; $\text{Enc}(k,m)=k\oplus m$, $\text{Dec}(k,c)=k\oplus c$; perfectly secret.

Prop 1.6 (key-length bound). $|\mathcal{K}|<|\mathcal{M}|\to$ no perfect secrecy. Indeed $|\text{supp}(\text{Enc}(K,m_0))|\le|\mathcal{K}|<|\mathcal{M}|$, so some m_1 lies outside the support and the two distributions differ. Allowing total-variation error ε gives **statistical secrecy** — useless once $|\mathcal{K}|\ll|\mathcal{M}|$.

Def 1.7 (scheme, v2). A **security parameter** λ replaces the message length: $k\leftarrow\text{Gen}(1^\lambda)$, messages in $\mathcal{M}_\lambda$, all algorithms p.p.t. in λ.

Def 1.8 (computational indistinguishability, IND). For all $m_0\neq m_1\in\mathcal{M}_\lambda$ output by the adversary and every p.p.t. distinguisher $D:\mathcal{C}\to\{0,1\}$,

$$\Pr[D(\text{Enc}(K,m_b))=b]\le 1/2+\text{negl}(\lambda),$$

with $b\leftarrow\{0,1\}$ chosen by the challenger.

Def 1.9 (semantic security). For every p.p.t. A there is p.p.t. S with

$$\Pr[A(\text{Enc}(K,M),g(M))=f(M)]\le\Pr[S(g(M))=f(M)]+\text{negl}(\lambda).$$

Thm 1.10 (Goldwasser–Micali). Def 1.8 $\Leftrightarrow$ Def 1.9. (If security fails for some m_1, then for infinitely many λ $\Pr[A(\text{Enc}(K,m_1))=f(m_1)]\ge\Pr[A(\text{Enc}(K,m_0))=f(m_0)]+1/\text{poly}$, contradicting IND; a distinguisher conversely breaks semantic security for M uniform on $\{m_0,m_1\}$.)

2 · Pseudorandom generators, stream ciphers

Def 1.11 (PRG). $G:\{0,1\}^\lambda\to\{0,1\}^{\ell(\lambda)}$ with $\ell(\lambda)>\lambda$ and $G(U_\lambda)\approx U_{\ell(\lambda)}$; $\ell(\lambda)-\lambda$ is the **stretch**.

Prop 1.12 (stretch amplification). A 1-bit stretch extends to any poly stretch. Write $G(s_i)=x_i\,\|\,s_{i+1}$ and $G'(s)=x_1\,\|\cdots\|\,x_n$, $n=\ell(\lambda)$. If G' is not a PRG, the hybrid argument yields a next-bit predictor $\Pr_{s}[P(x_i,x_{i+2},\cdots,x_n)=x_{i-1}]\ge\frac12+\frac{1}{\text{poly}(n)}$. But $x_i\cdots x_n$ is a function of s_i, so P predicts a bit of $G(s_{i-1})$ — contradicting that G is a PRG.

Prop 1.13 (stream cipher). If G is a PRG then $\text{Enc}(k,m)=G(k)\oplus m$, $\text{Dec}(k,c)=G(k)\oplus c$ is IND-secure. (For a supposed distinguisher D let $A(x)$ pick $b\leftarrow\{0,1\}$ and output $[D(x\oplus m_b)=b]$; then $\mathbb{E}[A(U)]=1/2$ while $\mathbb{E}[A(G(k))]\ge\frac12+\frac{1}{\text{poly}}$.)

A PRG with poly stretch XORed with an arbitrarily long message is a **stream cipher**.

3 · CPA security, PRFs, one-way functions

Def 2.1 (multi-message game). The adversary sends $(m_i^0,m_i^1)_{i\le l}$ with $m_i^0=m_i^1$ allowed, receives $c_i=\text{Enc}(k,m_i^b)$, wins if $b'=b$. Not adaptive.

Def 2.2 (CPA). Oracle access to $\text{Enc}(k,\cdot)$ before and after the challenge; only one challenge pair m^b must be distinguished. **Def 2.3 (multi-CPA).** Poly many pairs, each answered **instantly** with $\text{Enc}(k,m_i^b)$.

Prop 2.4. CPA $\Leftrightarrow$ multi-CPA. (Hybrid: answer the first j queries from m_i^0 and the rest from m_i^1.)

Deterministic encryption is never CPA-secure — query $m_i^0=m_i^1$.

Def 2.5 (PRF). $F:\{0,1\}^\lambda\times\{0,1\}^{n(\lambda)}\to\{0,1\}^{m(\lambda)}$ is p.p.t. computable and $F(k,\cdot)$ is indistinguishable from a uniformly random function. The **GGM tree** builds a PRF from a PRG; a PRF trivially gives a PRG.

CPA-secure randomized encryption. $\text{Enc}(k,m)=r_m\,\|\,(F(k,r_m)\oplus m)$, $r_m\leftarrow\{0,1\}^\lambda$. The r_m are distinct w.h.p.; conditioned on that, a large CPA advantage distinguishes F from a random function.

Def 2.6 (OWF). $f:\{0,1\}^*\to\{0,1\}^*$ is p.p.t. computable and for every p.p.t. A

$$\Pr[A(f(U_n),1^n)\in f^{-1}(f(U_n))]=\text{negl}(n).$$

Prop 2.6′. A PRG $G:\{0,1\}^\ell\to\{0,1\}^n$ with $n\ge2\ell$ is a OWF. (If A inverts w.p. $\ge1/n^c$, let $D(x)=1$ when $G(A(x,1^\ell))=x$ and a fair coin otherwise; then $\Pr[D(G(U_\ell))=1]\ge\frac12+1/n^c$ while $\Pr[D(U_n)=1]\le\frac12+2^{\ell-n}\le\frac12+2^{-n/2}$.)

Prop 2.7. A CPA-secure scheme yields a OWF: $f(x)=\text{Enc}(\text{Gen}(1^\lambda;x),0^\lambda)$.

Length-preserving padding is delicate. With $F:\{0,1\}^n\to\{0,1\}^{n/2}$ a OWF, let $F'(x)=x$ if $x=x'\,\|\,0^{n/2}$ and $F(x)$ otherwise. Then F' is still a OWF, but after padding to length n one has $F'(F'(x))=F'(x)$, so it is no longer one-way. (Pad with a 1 or with the length.)

4 · Hardcore bits, OWP, PRP

Def 3.1 (weak OWF, OWP). A **weak OWF** is p.p.t. and there is a fixed poly q such that every p.p.t. A inverts it on at most a $1-1/q(n)$ fraction of inputs (for large n). A **one-way permutation** is a OWF that is a permutation.

Def 3.2 (hardcore predicate). $h:\{0,1\}^*\to\{0,1\}$ with, for every p.p.t. A,

$$\Pr_{x\leftarrow\{0,1\}^\lambda}[A(1^\lambda,f(x))=h(x)]\le1/2+\text{negl}(\lambda).$$

Thm 3.3 (Goldreich–Levin). If a OWF exists, some OWF has a hardcore predicate. For a OWF f take $f_{\text{GL}}(x,y)=(f(x),y)$, $|y|=|x|=|f(x)|$, and the Hadamard bit $h(x,y)=\langle x,y\rangle$. A predictor with advantage $1/\text{poly}$ gives, for a noticeable fraction of x, $\Pr_{y|A_{\text{GL}}(f_{\text{GL}}(x,y))=\langle x,y\rangle}\ge\frac12+1/\text{poly}$; local list-decoding of the Hadamard code then recovers x. Pick a random $\le t$-dimensional subspace $V\le\mathbb{F}_2^n$; given correct values $\langle a,\cdot\rangle$ on V, majority over $x\in V$ of $g(x\oplus e_i)\oplus\langle a,x\rangle$ recovers $\langle a,e_i\rangle$. Values on V are correct when they are correct on generators $r_1,\cdots,r_t$ (probability 2^{-t}), and for $r_S=\bigoplus_{i\in S}r_i$ the pairwise independence plus Chebyshev gives error $\le1/((2^t-1)\varepsilon^2)$. With $t=\log(2n/\varepsilon^2)$ recovery is noticeable, so f is inverted. **Cor 3.4.** OWP p + hardcore $h\to$ PRG $G(x)=p(x)\,\|\,h(x)$.

Def 3.5 (universal OWF); Prop 3.6. Enumerate Turing machines and split $x=x_1\,\|\,x_2\,\|\cdots$ with $|x_i|=|x|/2^i$; then $f(x)=\text{TM}_1(x_1)\,\|\,\text{TM}_2(x_2)\,\|\cdots$ is a OWF whenever any OWF exists. (Padding gives a universal construction running in time $O(n^2)$.)

Thm 3.8 (Yao). Weak OWF $\to$ OWF. With f weak and $f'(x_1\,\|\cdots\|\,x_m)=f(x_1)\,\|\cdots\|\,f(x_m)$, $m=m(n)$ poly and independent of p: if A' inverts f' w.p. $\ge1/p(n)$, then either all blocks are "good" or some block is bad — but a bad block would be inverted by A', bounding the density of bad blocks; repetition amplifies the good ones. (A direct-product theorem; full proof in pset3.)

PRP / block cipher. $P:\{0,1\}^\lambda\times\{0,1\}^{n(\lambda)}\to\{0,1\}^{n(\lambda)}$ must be correct, with P and P^{-1} computable in p.p.t. **Secure** = a PRF; **strongly secure** = secure even with oracles to both P and P^{-1}.

Thm 3.7 (Feistel). With $x=x_0\,\|\,x_1$ and $x_{i+2}=F(k_{i+1},x_{i+1})\oplus x_i$, the 3-round network is a secure PRP and the 4-round network is a strong PRP. (Replace each F by a random f_i by a hybrid argument; collisions at x_2,x_3,x_4 occur with negligible probability by a union bound, after which the outputs are uniform and independent. Alternating directions require "local replacement"; see pset3.)

3 rounds are not strongly secure. Query (x_0,x_1), $(x_0\oplus r,x_1)$ and $P^{-1}(x_3^0,x_4^0\oplus r)$; then $x_3^1\oplus x_1^0=x_1^2\oplus x_3^0=f_2(x_2\oplus r)$, a relation a random permutation satisfies only with negligible probability. The cause: x_2 is reachable from both directions, while with 4 rounds cross-direction collisions such as $x_0^i\oplus f_1(x_1^i)=x_4^j\oplus f_3(x_3^j)$ are prevented.

5 · Modes of operation, block ciphers

ECB (Ex 4.1). $c_i=F_k(m_i)$ — unsafe: equal blocks leak equality. **CBC (Ex 4.2).** $c_0=\text{IV}$, $c_{i+1}=F_k(c_i\oplus m_{i+1})$; CPA-secure for a PRP F with a fresh uniform IV. **OFB (Ex 4.3).** $s_0=\text{IV}$, $s_i=F_k(s_{i-1})$, $c_i=m_i\oplus s_{i-1}$. **CTR (Ex 4.4).** $c_i=m_i\oplus F_k(\text{ctr}+i)$; needs a nonce unique per message; parallel and random-access.

SPN (Ex 4.5). $k=k_0\,\|\cdots\|\,k_m$; each round applies $\oplus k_i$ then a public permutation $P_i=L_i\circ S_i$ with S_i block-wise nonlinear and L_i linear. AES takes $S_i:x\mapsto x^{-1}$ in $\mathbb{F}_{2^8}$ (plus an affine map) and L_i a fixed matrix acting on 16 blocks. **Swap-or-not (Ex 4.6).** T rounds; in round i, if $F_k(\{x,x\oplus\Delta_i\})=1$ then $x\leftarrow x\oplus\Delta_i$, else x is unchanged. Generalizes to permutations of any set.

6 · CCA security, MACs, authenticated encryption

Def 4.7 (IND-CCA2). A gets 1^λ and oracles $\text{Enc}(k,\cdot)$, $\text{Dec}(k,\cdot)$; it outputs equal-length m_0,m_1; the challenger samples $b\leftarrow\{0,1\}$ and $c\leftarrow\text{Enc}(k,m_b)$; A keeps both oracles but may not decrypt c; it wins if $b'=b$. (Enc is randomized.) Earlier schemes fail: a slight modification of c still decrypts to a related message (malleability).

Def 4.8 (MAC, CMA). $\Pi=(\text{Gen},\text{Mac},\text{Vrfy})$, with $\Pr[\text{Vrfy}(k,m,\text{Mac}(k,m))=1]=1$; for deterministic MACs **canonical** verification checks $\sigma=\text{Mac}(k,m)$. The **chosen-message attack** gives oracle access to $\text{Mac}(k,\cdot)$ and asks for (m,σ) with m unqueried, winning if it verifies. **Strongly secure**: also allows a new σ' for an already queried m.

Def 4.9 (authenticated encryption). CPA-secure **and** unforgeable: a forger with an Enc oracle cannot output a ciphertext that decrypts to an unqueried message.

Prop 4.10. AE $\to$ CCA. Unforgeability means the adversary never needs a meaningful Dec query, so Dec can be removed and only the CPA game remains.

Prop 4.11. CPA-secure Enc + CMA-unforgeable Mac $\to$ encrypt-then-MAC is AE: $c'=\text{Enc}(k,m)$, $\sigma=\text{Mac}(k',c')$, $c=(c',\sigma)$. (CPA: replace k' by a random key, contrapose. Unforgeability: replace k by a random key, contrapose.)

Composition pitfalls. $c=(\text{Enc}(k,m),\text{Mac}(k',m))$ can fail CPA (the tag may leak m). MAC-then-encrypt ($c=\text{Enc}(k,m\,\|\,\sigma)$) fails CCA — e.g. a scheme that prepends a "trash bit" can be modified and

still decrypt validly. Only encrypt-then-MAC is safe.

CBC-MAC (Def 4.12). $\sigma_0 = 0^n$, $\sigma_i = F_k(\sigma_{i-1} \oplus m_i)$, $\sigma = \sigma_l$. Unforgeable only for **prefix-free** query sets; fix with ECBC-MAC by encrypting the last round with an independent key, $\sigma_l = F_{k'}(\sigma_{l-1} \oplus m_l)$, which is essentially random as long as the arguments $\sigma_{l-1} \oplus m_l$ are pairwise distinct. **Prop 4.13** ECBC-MAC is unforgeable (induction on the trie of the queried messages, as for Feistel).

NMAC (Ex 4.14). $H_k(M) = F(\cdots F(F(k, m_1), m_2)\cdots, m_l)$ and $\mathrm{Mac}(k \,\|\, k', M) = F(k', H_k(M))$. **PMAC (Ex 4.15).** $c_i = F_k(m_i \oplus L_i)$, $c = F_{k'}(\bigoplus c_i)$. **Ascon (Ex 4.16).** $S_{i+1} = p_k(S_i \oplus M_i)$.

7 · Hashing, random oracles

Def 5.1 (keyed hash). (Gen, H) with $\mathrm{Gen}(1^\lambda)$ giving $|k| = \lambda$; **collision resistant** if for every p.p.t. A

$$\Pr[H(k, m_0) = H(k, m_1) : A(1^\lambda) \to m_0 \neq m_1] = \mathrm{negl}(\lambda).$$

Universal hashing (Ex 5.2). $k \in \mathbb{F}_{2^\lambda}$, $H(k, p) = p(k)$ with $\deg p < 2^\lambda$ is collision resistant (a nonzero difference polynomial has fewer than 2^λ roots). The keyless notion is the same without k.

Prop 5.3. CRHF → OWF: inverting a random x to x' with $H(x') = H(x)$ gives $x' \neq x$ w.h.p., a collision. Converse is impossible in a black-box way, so one extends a fixed-length CRHF:

Merkle–Damgård (Ex 5.4). $H : \{0,1\}^{2\lambda} \to \{0,1\}^\lambda$ CR, $d_0 = \mathrm{IV}$, $d_i = H(d_{i-1}, m_i)$. Only equal-length messages remain a problem; reveal (or pad with) the length. **Merkle tree (Ex 5.5).** $x_p = H(x_l, x_r)$ over a binary tree of blocks; same length fix. **Weaker attackers.** Second preimage: given random x, find $x' \neq x$ with $H(x') = H(x)$. Preimage: given random y, find any x with $H(x) = y$.

Keyed hashing from a CRHF (Ex 5.6). $H(k \,\|\, m)$ suffers length extension; $H(m \,\|\, k)$ collides whenever $m \neq m'$ collide keylessly. **Hash-and-MAC:** key (k, s) and $\mathrm{Mac}'_{k,s}(m) = \mathrm{Mac}_k(H^s(m))$ with a fixed-length MAC and a seeded hash. **HMAC** is 2-nested: the prepend-keyed hash is applied again, so the key enters both layers.

Compression functions. Using a block cipher directly inside Merkle–Damgård is unsafe (weak against preimage attacks). **Davies–Meyer:** $H(x, y) = E_y(x) \oplus x$. Security needs an **ideal cipher**, not just a PRP, because the adversary controls the key. **Prop 5.9** Davies–Meyer is collision resistant under an ideal cipher $\mathcal{E}$: for distinct queries $i < j$ the value $m_i \oplus c_i = m_j \oplus c_j$ has probability $\leq 1/(2^n - j + 1)$ (randomness is fixed after query j), then union bound.

Random-oracle model. Prop 5.7 A random oracle is a PRG against any A^H making polynomially many queries — no computational bound needed: given $y = H(x)$, w.h.p. A never queried x, so the real world degenerates to the ideal one. **Prop 5.8** A random oracle is collision resistant: the output pair must appear among the queries, reducing to the non-adaptive bound.

Def 5.10 (indifferentiability). C^F is indifferentiable from P if some simulator S gives, for every A, $|\mathbb{E}[A^{C^F, F}] - \mathbb{E}[A^{P, S^P}]| = \mathrm{negl}(n)$. Needed because the adversary may access the underlying primitive F. A 5-round Feistel network is not indifferentiable from an ideal permutation; 10 rounds suffice.

8 · Public-key cryptography

Def 6.1 (key agreement). A pair of interactive algorithms $(A(1^\lambda), B(1^\lambda))$, **correct** (identical outputs) and **passively secure**: $(T, k) \approx (T, k')$ with T the transcript and k' uniform — so k is hard to find from T.

Merkle puzzles (Ex 6.2). Alice sends N puzzles each costing $\Theta(N)$ to solve; Bob solves one ($\Theta(N)$), the eavesdropper must solve all ($\Theta(N^2)$).

Def 6.3 (PKE). $\mathrm{Gen}(1^\lambda) \to (pk, sk)$, $\mathrm{Enc}(pk, m)$, deterministic $\mathrm{Dec}(sk, c)$ with $\mathrm{Dec}(sk, \mathrm{Enc}(pk, m)) = m$. **Def 6.4 (CPA).** A receives pk, sends m_0, m_1, receives $\mathrm{Enc}(pk, m_b)$, outputs b'; no encryption oracle is needed since pk is public. PKE → key agreement: Bob publishes pk, Alice sends $\mathrm{Enc}(pk, k)$.

Def 6.5 (KEM). $(\mathrm{Gen}, \mathrm{Encaps}, \mathrm{Decaps})$ with $\mathrm{Gen}(1^\lambda) \to (pk, sk)$, $\mathrm{Encaps}(pk) \to (k, c)$, $\mathrm{Decaps}(sk, c) = k$; secure if k is indistinguishable from uniform given pk and c.

Def 6.6 (trapdoor permutation). $(\mathrm{Gen}, F, \mathrm{Inv})$: $\mathrm{Gen}(1^\lambda) \to (f, \mathrm{tp})$ where $f = F_{\mathrm{pp}}$ is injective with public parameter pp; $\mathrm{Inv}(\mathrm{pp}, \mathrm{tp}, f(x)) = x$; f is one-way without tp.

RSA (Ex 6.7). $N = pq$, $ed \equiv 1 \pmod{\varphi(N)}$, $\mathrm{pp} = (N, e)$, $\mathrm{tp} = d$; $F_{\mathrm{pp}}(x) = x^e \bmod N$, $\mathrm{Inv}(\mathrm{pp}, \mathrm{tp}, y) = y^d \bmod N$, $x \in \mathbb{Z}_N^\times$.

Key agreement from a TDP (Def 6.8). Alice runs Gen and sends pp; Bob sends $y = f(x)$; Alice outputs $h(\mathrm{Inv}(\mathrm{pp}, \mathrm{tp}, y))$, Bob outputs $h(x)$. Given $f(x)$, x is easily distinguishable from random, so h must be a random oracle (the adversary would have to query x) or a hardcore predicate (one pseudo-bit per sample, combined by extractors — the notes call this handwaving).

Prop 6.9. With a hardcore $h(x, r)$ for $(f(x), r)$, the following is strongly secure: Bob sends $(y = f(x), r)$; both output $h(x, r)$. A distinguisher with advantage ε inverts f: on input y, sample r, run it on (y, r), compare with $h(x, r)$.

Assumptions. (0) RSA. (1) RSA with fixed e. (2) Strong RSA: given y, find (x, e) with $e \neq \pm 1$ and $y \equiv x^e$. (3) λ-bit safe primes $p = 2p' + 1$. (4) Factoring is hard. (5) Discrete log: from g, g^x find x. (6) CDH: from g, g^x, g^y find g^{xy} — gives a key agreement with key $h(g^{xy})$. (7) DDH: $g^{xy} \approx g^z$, after which h is unnecessary. Implications: $7 \to 6 \to 5$; $2 \to 0$; $2 \to 1$.

ElGamal (Def 6.10). Key $pk = (G, g, g^x)$, $sk = x$. Under DDH $\mathrm{Enc}(m) = (g^y, g^{xy} m)$ and $\mathrm{Dec}(t, c) = t^{-x} c$; under CDH $\mathrm{Enc}(m) = (g^y, H(g^{xy}) \oplus m)$ and $\mathrm{Dec}(t, c) = c \oplus H(t^x)$.

Prop 6.11 (worst case to average case). If RSA with fixed (N, e) is hard, it is hard on random instances: randomize $y' = yr^e$ (for discrete log, $h' = hg^r$), invert y' to z, then $x = zr^{-1}$ (resp. $x = z - r$).

Keyed CRHF from discrete log. Let $|G| = p$ be prime, $g \in G$, $h = g^s$; then $H(x, y) = g^x h^y$ — a collision reveals s (hence the discrete log). Instantiate in $\mathbb{QR}_p$: for $p = 2p' + 1$ the quadratic residues form a cyclic group of order p'.

Def 6.12 (homomorphic encryption). If $c = \mathrm{Enc}(f(x_1, x_2))$ can be computed directly from $c_1 = \mathrm{Enc}(x_1)$, $c_2 = \mathrm{Enc}(x_2)$, the scheme is homomorphic. Such schemes are never CCA-secure.

9 · Notions at a glance

Perfect secrecy — $\mathrm{Enc}(K, m_0) \equiv \mathrm{Enc}(K, m_1)$; needs $|\mathcal{K}| \geq |\mathcal{M}|$; OTP attains it. **Statistical secrecy** — the two distributions differ by total variation ε. **IND (computational)** — $\Pr[D(\mathrm{Enc}(K, m_b)) = b] \leq \frac{1}{2} + \mathrm{negl}$; equivalent to semantic security (GM). **CPA** — plus an encryption oracle; equivalent to its multi-message form; deterministic schemes excluded. **CCA2** — plus a decryption oracle, except on the challenge ciphertext. **AE** — CPA-secure and unforgeable; implies CCA; encrypt-then-MAC attains it. **MAC** — CMA unforgeability (strong: no new tag for an old message). **Hash** — collision resistance; a CRHF is a OWF, while second-preimage and preimage are weaker attacker goals. **PKE / KEM** — the CPA game with pk public; a KEM encapsulates a random k. **Key agreement** — correct and $(T, k) \approx (T, U)$; Merkle puzzles give a $\Theta(N^2)$ work gap.

10 · Reduction recipes and pitfalls

Recipes. Hybrid argument — swap one object at a time (multi-CPA ⇔ CPA, stretch amplification, Feistel round replacement). **Distinguisher from inverter** — guess and test: $D(x) = 1$ iff $G(A(x)) = x$ (PRG → OWF). **Inverter from distinguisher** — embed the challenge: $A(x) = [D(x \oplus m_b) = b]$ (stream cipher), local list-decoding (GL). **Union bound on collisions** — Feistel, Davies–Meyer and ECBC-MAC need pairwise distinct arguments. **"The query must be made"** — in the ROM a value never queried looks uniform (RO → PRG; TDP + RO key agreement). **Contraposition with a random key** — the encrypt-then-MAC proof. **Direct product** — Yao's weak-to-strong amplification.

Pitfalls. A deterministic scheme is never CPA-secure; CBC needs a fresh uniform IV and CTR a unique nonce; CBC-MAC is safe only for prefix-free queries (use ECBC); $H(k \,\|\, m)$ suffers length extension whereas $H(m \,\|\, k)$ inherits keyless collisions; a 3-round Feistel network is not a strong PRP; padding a OWF must preserve one-wayness; ElGamal is malleable and homomorphic encryption is never CCA-secure; ROM proofs hold only in the ROM, and indifferentiability is needed once the adversary may call the underlying primitive.

11 · Constructions at a glance

Construction	From	Gives
one-time pad	uniform k	perfect secrecy
stream cipher	PRG	IND
$r \,\|\, F(k, r) \oplus m$	PRF	CPA
CBC/OFB/CTR	PRP	CPA, fresh IV
Feistel, 3/4 rounds	PRF	PRP, strong
GGM tree	PRG	PRF
$p(s) \,\|\, h(s)$	OWP + HC	PRG
swap-or-not	PRF	PRP, any set
CBC-MAC, ECBC	PRF	CMA
encrypt-then-MAC	CPA + MAC	AE → CCA
MD, Merkle tree	fixed CRHF	CRHF
universal hash	polynomial	CRHF
Davies–Meyer	ideal cipher	CRHF
hash-and-MAC	CRHF + MAC	MAC, long msg
RSA, ElGamal	RSA / DDH	PKE
TDP + HC/RO	TDP	KA, KEM
Merkle puzzles	none	KA, $\Theta(N^2)$ gap

12 · Implication map

weak OWF → OWF (Yao) → OWF with a hardcore bit (GL); OWP + hardcore → PRG (Cor 3.4) → PRF (GGM) → PRP (Feistel) → block cipher and its modes; PRG with $n \geq 2\ell$ → OWF; CPA-secure Π → OWF; CRHF → OWF; random oracle → PRG and CRHF; fixed-length CRHF → arbitrary-length CRHF (Merkle–Damgård, Merkle tree); CPA + CMA-unforgeable MAC → encrypt-then-MAC → AE → CCA; TDP + (random oracle or hardcore bit) → key agreement; DDH → ElGamal, CDH → key agreement; PKE → key agreement; homomorphic encryption → not CCA-secure.